Project Proposal for Deep Learning Artificial Image Detection

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Abstract

With the rapid growth of generative AI, it's difficult to distinguish between real images and generated ones. This project investigates whether training a deep-learning detector on multiple image generators can improve its ability to identify images from completely unseen generators. We will use the GenImage dataset, which includes real images and synthetic images generated by image generators like Stable Diffusion, ADM, GLIDE, VQDM, Wukong, BigGAN, and Midjourney. We will compare a conventional image classifier with a pre-trained vision-based model using metrics such as accuracy, F1-score, and AUROC. The detector will also be evaluated on a public face-manipulation dataset to examine its ability to identify traditional deepfakes. Finally, its performance will be compared with zero-shot predictions from multimodal LLM's such as GPT-5.6 and Claude Opus. This study aims to understand how generator diversity improves detection and generalization.

1 Introduction

Currently, there are many AI-generated images that are often passed off as authentic. While sometimes such images are used for art and design, other times they are used to spread false information. Images can be doctored, or an entirely fallacious image can be generated with a few words to present inaccurate information. This project aims to train an AI-generated image detector on multiple generators to determine whether the detector can generalize detection for AI-created images or if it learns the patterns of a specific generator. As deepfakes are becoming increasingly easier to create, it is important to determine if we can improve detectors of AI-created images. By using a larger training subset consisting of images by different generators, this project attempts to determine whether we can improve the detection of synthesized images.

1.1 Why it is Useful

As AI becomes increasingly popular, it is simpler and faster to create image generators that can fool AI detectors. Existing detectors become outdated as AI models begin to use updated patterns to generate images that are designed to deceive current detectors that are only trained on older generators. The significance of our project is that it will determine whether it will improve the accuracy of AI detectors if they are trained to find general patterns between image generators with a dataset that consists of images from multiple AI models instead of making a separate dataset for each model. By creating this project, we are aiming for a more reliable detector that will catch more images created by newer models.

1.2 Why This is Interesting

It is interesting to determine what an AI detector relies on to determine whether an image is authentic or if it has been generated by an AI model. There may be patterns or textures in the image that suggest it was created by a specific model, or a model may have a certain resolution for its images. By researching whether we can improve an AI detector by training it on multiple image generation AI models, we are comparing the performance of different image generators and evaluating whether we can enhance the accuracy and effectiveness of AI detectors.

2 Potential Solutions

We will treat this as a binary image-classification task in which a detector predicts whether an image is real or AI-generated. In its cross-generator experiments, GenImage trained separate detectors using one generator at a time and then tested them on other generators. We propose extending this approach by training a single detector on images from multiple generators while leaving one generator completely unseen for testing. We will vary the number of training generators using ($k=\{1,2,4,7\}$) and compare a standard CNN with a transformer-based model. This will test whether greater generator diversity helps the models learn general signs of AI generation instead of generator-specific artifacts.

Domain-Adversarial Training: We will treat each image generator as a different domain and train the model to detect real versus AI-generated images while reducing its reliance on generator-specific features. A gradient reversal layer can help the model learn features that are common across different generators. This may improve its ability to detect images from generators it has not seen during training.

Confidence and Calibration: Along with accuracy, precision, recall, and F1-score, we will also look at how confident the detector is in its predictions. We expect the model to be less confident when it sees images from generators it has not encountered during training. Calibration can help us check whether the model's confidence matches how often its predictions are actually correct. This gives us a better idea of how reliable the detector would be in real-world situations. During training, we will also focus on image augmentations such as compression, blur, resizing, and image noise to evaluate their impact on cross-generator detection and model robustness.

Contrastive Learning: Contrastive Learning.

We will explore contrastive learning to help the model recognize similarities between AI-generated images from different generators. For example, images from BigGAN and Stable Diffusion can be treated as similar examples, while real photographs are treated as different. The goal is to group AI-generated images together and separate them from real images, helping the detector learn general patterns instead of relying on artifacts from a specific generator. As a possible extension of this project, we will explore whether AI-generated image detection models can also work on deepfake datasets. This will help us understand whether the features learned from images generated by different AI models can be applied to other types of manipulated images.

3 Expected Outcomes

We expect that training a detector on images from multiple AI generators will help it perform better on generators that it has not seen during training. If this is true, it would show that the detector is learning features that are common across AI-generated images instead of only learning the specific artifacts of one generator. We will compare models trained with different numbers of generators to see if increasing the diversity of the training data improves performance on unseen generators.

We will be comparing different image classification approaches, including a CNN and a transformer-based model, to see how their performance differs when detecting AI-generated images. Furthermore, we will compare our trained models with zero-shot approaches, such as ChatGPT and Claude, to see how they perform on the same classification task.

For the robustness experiments, we expect that changes like compression, blur, resizing, and noise could potentially affect detection performance. We will test whether training with these augmentations can help to make detector performance better on images that have been modified or reduced in quality.

As an extension, we plan to test whether our detector can also identify deepfake images. Since our main experiments focus on images generated by different AI generators, this would allow us to see whether the features learned by the detector can generalize to a different type of image manipulation. Overall, our goal is to determine whether using multiple generators during training can make AI-generated image detection more generalizable and robust.

4 Timeline

4.1 Week 1 - 2 Research and Data Preparation

Research previous works on AI-generated image detection. Collect datasets of real images and fake subsets, preprocess data, and create balanced, generator-based splits.

4.2 Week 3 - 5 Multi-Generator Training

Train baseline detectors on images from 1 generator vs. incremental combinations.

4.3 Week 6 - 9 Experiments

Evaluate performance of multi-generator classifiers and LLMs (ChatGPT and Claude). Expand testing to deepfakes if time permits.

4.4 Week 10 - 12 Analysis and Report

Generate visualizations and cross-generator performance heatmaps. Compile results and write the final paper.

References

Yongqi Yang, Zhihao Qian, Ye Zhu, Olga Russakovsky, and Yu Wu. 2025. Scaling Up Deepfake Detection by Learning from Discrepancy

M. Zhu, H. Chen, Q. Yan, X. Huang, G. Lin, W. Li, Z. Tu, H. Hu, J. Hu, and Y. Wang, "GenImage: A Million-Scale Benchmark for Detecting AI-Generated Image," *Advances in Neural Information Processing Systems*, vol. 36, 2023.